Element H

Prototype Testing and Data Collection Plan

As a team we will know if the solution works when asking our stakeholders if the solution satisfies all their user needs. This could be accomplished by asking more experts in the field of hydroponics to evaluate our design and give us feedback on what is good, and what needs improvement. Adding to this we as a team can also self-evaluate within groups and share within the team what we should change, and what individuals liked. Not only would this better prepare us for presentation, but it will also help us as a group understand the amount of revisions needed to produce great products. We plan to test the prototype by finding a website that would allow us to animate the water to detect any leaks or any splashing that Mr. Roberson spoke about. Another way we could test it in future projects is actually building the design since due to time constraints we could not this time. Another way we can test our prototype is by getting it evaluated by many more trusted sources. The optimal results for this project is a 3D design that fulfills the user needs and is approved by our stakeholders and multiple experts. Adding to this the design should also be easy to understand by our team and anyone who uses it to create their own hydroponics unit. The minimum required result is that it is approved and is justified by multiple experts like said in the sentences before. We will test the viability of our design with regard to growth rates, sustainability, as well as efficiency by measuring the amount of growth in the different plant variety, and how much the design has broken down with a goal of at least a two year imperishability and a constant rate of growth as the minimum for success. In order to complete this project we have to present our 3D model to multiple stakeholders and experts which addresses any concerns our team has about our design faltering. Additionally, we will have to look over the design as a group giving feedback to one another to ensure that all members agree with the final design.